Esau Hervert
MATH 6313

Homework 1.1

Problem 2.1

Write a program to compute the value of the machine number ϵ.

```
public class problem21 {

	public static void main(String[] args){
		float x = 1; // This will be the number that we keep cutting in half
until we get our epsilon.
		int k = 0; // This will be the variable we use to index our cuts.


		System.out.print("Single Precision\n");

		while ((1+x)>1) // This loop will cut x until it gets too small to make
a difference.
		{
			// Displaying the k value and the result of 1 + x.
			System.out.print("K: " + k + "    " + "1 + x = " + (1 + x) +
"\n");

			x=x/2; // Cut the number in half.
			k=k+1; // Increase the index count.
		}

		// The k value that causes 1 + x to be indistinguishable from 1 will be
the k value that comes after
		// the last k value that does make 1 + x not equal 1, thus 2^(k-1) will
be the machine number in question.
		System.out.print("\nAt K = " + k + ", we get that 1 + x = 1, thus the
machine epsilon for "
					+ "single precision is 2^(-" + (k - 1) + ").\n\n");

		System.out.print("Double Precision\n");

		// Reseting everything for the double case.
		k = 0;
		double X = 1; // The only change is that X is now a double.
		while ((1+X)>1)
		{
			System.out.print("K: " + k + "    " + "1 + x = " + (1 + X) +
"\n");

			X=X/2;
			k=k+1;
		}

		System.out.print("\nAt K = " + k + ", we get that 1 + x = 1, thus the
machine epsilon for "
					+ "double precision is 2^(-" + (k - 1) + ").\n\n"); }

}
```

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Results:

```
Single Precision
K: 0   1 + x = 2.0
K: 1   1 + x = 1.5
K: 2   1 + x = 1.25
K: 3   1 + x = 1.125
K: 4   1 + x = 1.0625
K: 5   1 + x = 1.03125
K: 6   1 + x = 1.015625
K: 7   1 + x = 1.0078125
K: 8   1 + x = 1.0039062
K: 9   1 + x = 1.0019531
K: 10  1 + x = 1.0009766
K: 11  1 + x = 1.0004883
…
K: 20  1 + x = 1.000001
K: 21  1 + x = 1.0000005
K: 22  1 + x = 1.0000002
K: 23  1 + x = 1.0000001

At K = 24, we get that 1 + x = 1, thus the machine epsilon for single precision is
2^(-23).
```

```
Double Precision
K: 0   1 + x = 2.0
K: 1   1 + x = 1.5
K: 2   1 + x = 1.25
K: 3   1 + x = 1.125
K: 4   1 + x = 1.0625
K: 5   1 + x = 1.03125
K: 6   1 + x = 1.015625
K: 7   1 + x = 1.0078125
K: 8   1 + x = 1.00390625
K: 9   1 + x = 1.001953125
K: 10  1 + x = 1.0009765625
K: 11  1 + x = 1.00048828125
…
K: 47  1 + x = 1.000000000000007
K: 48  1 + x = 1.0000000000000036
K: 49  1 + x = 1.0000000000000018
K: 50  1 + x = 1.0000000000000009
K: 51  1 + x = 1.0000000000000004
K: 52  1 + x = 1.0000000000000002

At K = 53, we get that 1 + x = 1, thus the machine epsilon for double precision is
2^(-52).
```

Is this the exact value or the approximation of it?

This is the exact value because the machine operates on base 2 and since we started with $1 = 2^0$ and kept cutting it in half, our result was a binary number that could not be cut anymore thus we got that the number we found was the smallest number that we could have on the machine.